Week 3 Lecture Notes

Finding, Reading, and Evaluating Literature

EDCI 59100-016 | Summer 2026

Week 3 Overview

This week moves from planning a search to running, documenting, screening, and evaluating it. The goal is to build a transparent search trail, identify a preliminary set of candidate studies, and begin judging relevance and quality in relation to the selected literature review methodology.

Learning Objectives

1. Implement a search strategy using discipline-appropriate databases and structured search strings based on the literature review plan.
2. Document the search process transparently in a search log, with the level of structure aligned to the chosen review type.
3. Screen and evaluate candidate articles for relevance and quality, making initial include, maybe, and exclude decisions with justification.
4. Organize sources using reference management software to support later synthesis and writing.

Recommended Weekly Flow

1. Search within databases for the topic.
2. Read and evaluate candidate literature.
3. Explore reference management software and citation managers.
4. Evaluate AI tools through algorithmic thinking for literature review work.
5. Complete the discussion on screening decisions and search challenges.
6. Complete the Search Results and Source Evaluation Matrix and the Final Literature Review Plan.

Search Strategies and Databases

Searching Within Databases for Your Topic

Phase 1: Purpose

What's different this week... Last week, you learned how searching works. This week, you will apply it to your own literature review. • Use your research question(s) • Test your keywords • Apply decisions based on your review methodology Key shift: You are building and refining your search process.

Before you begin, gather:

• Your research question(s) or purpose • Your keywords • Your review type • Your initial search plan You are testing and improving your plan.

Phase 2: Build Your Search Strategy

In this phase, you will build and test your search strategy step by step.

2A. Choosing the Right Database

Different topics require different databases. • Education → ERIC, PsycINFO • Technology → IEEE, ACM • Health → PubMed • Broad → Scopus, Google Scholar Key idea: The "best" database depends on your field and topic. How do you know which databases you should use... • Start with the databases in your Initial Literature Review Plan • Ask Yourself: Where is research in my field typically published... Does this database include relevant journals... • Look at: Purdue library subject guides - Link to Library Guides Databases linked in articles you've found • If unsure: Use one discipline-specific database Compare with Google Scholar You are testing databases—not choosing a final one.

2B. Build 2–3 Search Strings

This means: • If your search is too simple → too many unrelated results • If your search is too narrow → you may miss important studies Search strings help you control and refine your results. How to build a search string: • Identify 2–3 key concepts from your research question • List synonyms for each concept • Combine them using: AND → connects different concepts OR → connects similar terms • Small changes in wording will produce different results. You are testing. You are not finalizing. Example Search Strings:

How Databases Read Your Search

Databases match exact words and combinations—not natural language. • Boolean operators must be in ALL CAPS • Quotation marks keep phrases together • Parentheses control grouping • Small changes in formatting can significantly change results. Embedded media/resource: YouTube video player

How to Use Your Search String

• You can paste your full search string into a database search box. • Or, use advanced search and split the string across multiple boxes.

How This Looks in a Database

Concept / Term Group 1	**Operator**	**Concept / Term Group 2**
AI OR "artificial intelligence"	AND	feedback

Begin identifying inclusion and exclusion criteria

Before you review your search results, begin thinking about what counts as a relevant source. These are your initial inclusion and exclusion criteria. These criteria will guide your Keep / Maybe / Exclude decisions. Based on your topic and review method, consider: • What types of sources you want to include • What types of sources you may exclude • What characteristics matter (e.g., population, context, timeframe, method) You will refine these as you search and evaluate sources. Next, consider how your review methodology influences your search decisions.

Phase 3: Apply to Your Method

Phase 4: Practice the Process

Try It: Compare and Refine Your Search (10 minutes)

You are not aiming for a specific number of articles. Instead, focus on whether your sources: • Consistently relate to your topic • Represent your method (e.g., variation, coverage, key works) • Begin to show patterns across studies

Step 1: Choose ONE search string from your list Step 2: Run TWO searches

• A university database • Google Scholar

Step 3: Compare results Look at the first 5–10 results:

• What types of sources appear... • Are they similar or different... • Which is more useful for your topic... Use your inclusion and exclusion criteria from the searching activity to guide your decisions. You may refine these as you read more closely. Step 4: Make ONE change • Change a keyword • Add specificity • Refine your focus Run it again. Did it improve...

Now apply this to YOUR method

• Narrative Review: Are these studies key and influential... Do they help explain the topic clearly... • Integrative Review: Do these results include different types of studies (quantitative, qualitative, theoretical)... Or are they too similar... • Theoretical Review: Are you finding concepts, frameworks, or foundational ideas, or mostly empirical studies... • Scoping Review: Do your results reflect a range of contexts, populations, or study types... • Systematic Review: Can you clearly justify how these results would be included or excluded based on defined criteria... Based on your method: What do you need to adjust in your search to better match your goal...

Phase 5: Document & Reflect

Instructions:

This week, you will use one working document across several activities and use it for your discussion and assignments. Working Searching and Evaluating Literature Document This document will help you track your searches, screening decisions, and notes as you move between searching and evaluating sources. Start by completing the Search Log and Initial Screening sections in the document (See the Sheets at the bottom of the document). You will continue adding to this document in the next activity.

Search Log

No.	Database	Search String	Filters	Results	Notes
1	ERIC	(“artificial intelligence” OR AI) AND feedback AND writing	Peer-reviewed; last 10 years	1,245	Too broad; includes K–12 and higher ed; results are not well targeted.
2	ERIC	(“artificial intelligence” OR AI) AND feedback AND writing AND “higher education”	Peer-reviewed; last 10 years	312	More focused; results align with the target population.
3	Google Scholar	Use a comparable search string for cross-checking coverage.	Adjust filters as available.	Record count	Compare whether results are broader, narrower, or more useful than database results.

Initial Screening Notes

For each source you find, think about whether it aligns with your topic and your review type. Use a screening process that connects the source decision to the method you are using:

- Narrative: explains the topic.
- Integrative: adds variation across study types or contexts.
- Theoretical: contributes concepts, frameworks, or foundational ideas.
- Scoping: adds a new dimension to the broader literature landscape.
- Systematic: meets predefined inclusion criteria.

Article Title	Decision	Why	Fits Method
AI Feedback in College Writing Courses (2022)	Keep	Direct match to topic; focuses on higher education and provides empirical findings.	Fits an integrative review because it provides empirical evidence to compare across studies.
AI Tools for K–12 Writing Support (2019)	Maybe	Relevant topic but different population; may be useful for comparison.	Fits an integrative review by adding variation across contexts.

After screening articles, your decision will depend on your review methodology.

Examples: How "Fits Method" Changes by Review Type

Notice how the same topic is evaluated differently depending on the review methodology.

Review Type	Article Title	Decision	Fits Method
Narrative	AI Feedback in College Writing (2022)	Keep	Explains key ideas clearly; widely cited and relevant to the topic.
Integrative	AI Feedback in College Writing (2022)	Keep	Provides empirical evidence to compare with other study types.
Theoretical	AI Feedback in College Writing (2022)	Maybe	Useful only if it connects to conceptual or theoretical discussions.
Scoping	AI Feedback in College Writing (2022)	Keep	Represents one context, higher education, within the broader literature landscape.
Systematic	AI Feedback in College Writing (2022)	Keep / Exclude	Include only if it meets predefined criteria such as population, design, and timeframe.

What comes next

You have now:

- Built and tested your search strategy.
- Started identifying inclusion and exclusion criteria.
- Begun making initial screening decisions.

In the next activity, Reading & Evaluating Literature, you will continue this process by reading sources more closely, refining your decisions, and identifying patterns in the literature.

Reading & Evaluating Literature

This activity builds directly on your search work—you will continue using the same Searching and Evaluating Literature working document. You will continue using your search results to: • Read sources more efficiently • Apply and refine your inclusion/exclusion criteria • Make more confident screening decisions • Start identifying patterns in the literature

What changes in this stage...

You are shifting from collecting → evaluating. You may still find and add new articles, but your focus is now on deciding: • Which sources are relevant • Which sources are useful • Which sources you will include in your review • How you decide to keep or exclude sources • What patterns you are starting to notice in the literature You will move back and forth between searching and evaluating as you refine your literature.

How this activity fits in your process

In this activity, you focus on reading and evaluating sources you have already found. This complements your searching work by helping you: • Decide which sources to keep, revisit, or exclude • Justify your screening decisions • Identify early patterns in the literature The next step is learning how to read and evaluate sources efficiently.

How to Read & Evaluate Sources Efficiently

Reading Sources (Quickly and Strategically)

You do NOT need to read every article fully. Instead, use a quick screening process.

Step 1: Title & Abstract

• Read the title and abstract first • Ask: Is this related to my topic... The abstract gives a summary of the whole study

Step 2: Skim the Article

• Read the introduction and conclusion • Look at methods and results • Scan headings, tables, and key terms Skimming helps you quickly decide if a source is worth reading in detail

Step 3: Make a Quick Decision

• Keep → clearly useful • Maybe → unclear, review later • Exclude → not relevant Use your inclusion and exclusion criteria to guide your Keep / Maybe / Exclude decisions. You may refine these criteria as you read more closely. Rule: If unsure, keep it for now. Resources: How to Read A Scholarly Article – Key Reading Strategies - Dennis Learning Center Mastering Skimming Articles for Effective Research | Nova Scholar Education

Quick Source Notes

Continue working with the Searching and Evaluating Literature document to track searches, screening decisions, and notes as you move between searching and evaluating sources. Use a structure like the table below as you read articles.

Source	**Main Idea**	**Method**	**Key Finding**	**Notes / Why It Matters**

Selected article or source	Summarize the central topic or argument.	Identify study design, review type, or conceptual approach.	Record the most relevant finding or claim.	Explain how the source supports, complicates, or does not fit your review.

Matrix assignment. Embedded media/resource: YouTube video player

Evaluating Sources

When you first encounter a potential source, you'll want to know very quickly whether it is worth reading in detail and considering for your literature review. To avoid wasting time on unhelpful sources as much as possible, it's generally best to run each article, book, or other resource you find through a quick checklist, using information you can find by skimming through the summary and introduction. The two most common forms of early source evaluation are the "Big 5 Criteria" and the "CRAAP Test." These cover the same most significant variables for evaluation, and which one to use comes down to preference. Big 5 Criteria: The most important criteria for evaluating a potential resource are: • Currency: When the source was published • Coverage/Relevance: How closely related the source is to your topic and research question • Authority: Who wrote the source and whether they are likely to be credible on the subject • Accuracy: Whether the information is accurate or not (this will be more heavily evaluated further on in the research process, but you should skim through quickly for any obviously inaccurate information as a means of disqualifying the source) • Objectivity/Purpose: Whether or not the source presents a biased point of view or agenda Helpful questions for initial evaluation: • When was this source published... • Is this source relevant to your topic... • What are the author or authors' qualifications... • Is the resource scholarly/peer-reviewed... • Are sources cited to support the author's claims... • Does the website or journal the source comes from have a bias to their reporting... • Do you notice biased or emotional language in the summary or introduction... • Do you notice spelling or grammatical errors in a quick examination of the source... Source: Evaluating Sources - Conducting a Literature Review - Library Services at Albany College of Pharmacy and Health Sciences;

What makes a source useful for Your research...

• Relevance: Does it match your topic... • Authority: Is the author credible... • Evidence: Are claims supported... • Purpose: Is it research-based (not opinion)... Strong sources are both reliable and appropriate for your research Reminder: You are not aiming for a specific number of articles. Instead, focus on whether your sources: • Consistently relate to your topic • Represent your method (e.g., variation, coverage, key works) • Begin to show patterns across studies

Now apply this process to your own search results.

Try It: Screen and Select (10 minutes)

As you work through this activity, pay attention to: • Your search successes and challenges • Your inclusion and exclusion criteria • Your keep / maybe / exclude decisions • Patterns you are starting to notice in the literature Be specific in your reasoning—this is the same logic you will use in your literature review plan. The patterns you begin to notice now will later help you identify themes, patterns, and tensions in your literature review. These patterns will help you group studies and organize your literature review. Step 1: Take one of your search results lists Step 2: Quickly review 10 articles • Read titles and abstracts only • Do NOT read full articles

Step 3: Make decisions Mark each as:

• Keep → clearly useful and fits your review • Maybe → unclear or needs deeper reading • Exclude → not relevant to your topic As you make decisions, explain how each source meets (or does not meet) your inclusion criteria. "Maybe" articles are sources you are unsure about. You may revisit these after reading more or re- fining your criteria. Step 4: Select three "Keep" articles Step 5: How does it fit your review method (narrative, integrative, theoretical, scoping, or systematic)... • Why is this relevant... • How does it fit your method... Goal: Practice making fast, justified decisions.

Reference Management Software (RMS) Citation Managers

Literature source management software—often called Reference Management Software (RMS) or citation managers—is highly relevant and more powerful than ever. Beyond simply storing citations, these tools are now full research ecosystems designed to save time, organize PDFs, and format bibliographies dynamically.

Core Advantages

• Instant Formatting: Automatically generate and switch between citation styles (e.g., APA to IEEE) with a single click. • Workflow Automation: Capture paper details and PDFs directly from your browser and integrate with tools like Microsoft Word and Google Docs. • Collaboration: Create shared libraries to collaborate with others on research projects.

Modern Enhancements

Today's RMS platforms are no longer just static repositories. Tools like Zotero combine tagging, highlighting, and annotation extraction, while newer AI-enabled tools incorporate semantic search and literature synthesis into your workflow.

What to Use

• Zotero: Free, open-source, highly customizable with strong browser integration. • EndNote: Industry standard for advanced academic and large-scale research projects. • Mendeley: Popular free tool with strong PDF management and annotation features. Researchers generally agree that while building small bibliographies manually is possible, relying on a citation manager is essential for scaling research projects, preventing plagiarism, and avoiding formatting errors.

How These Tools Support Your Workflow

Workflow-based process: Import → Organize → Attach PDFs → Annotate → Cite while writing Now try using one of these tools with your own sources.

Try It: Explore a Reference Manager (10–15 minutes)

Choose ONE tool to explore: • Zotero • Mendeley • EndNote Complete the following: • Install or open the tool • Add at least 2–3 sources from your current search results • Explore one feature (e.g., tagging, organizing, or attaching PDFs) Use sources from your Search Results & Source Evaluation Matrix for this activity. As you work, consider: • How might this tool help you stay organized... • What feature seems most useful for your workflow... • What questions do you have about using this tool... You are not expected to master the tool—just begin exploring how it could support your research process. You may choose to continue using one of these tools throughout the course to organize your sources.

AI Tools - AI Bias Activity

AI Tools - AI Algorithmic Thinking for Literature Review

Overview

This lesson introduces key ideas about how AI systems generate, organize, and prioritize information—and how these processes can influence your literature review searching and evaluation. As you work through this page: • Pay attention to how AI tools shape what information becomes visible or emphasized • Consider how bias, incomplete information, or lack of transparency could affect your research deci- sions • Think about how you will evaluate and use AI tools as part of your literature review process What you will do next: • Apply these ideas by evaluating multiple AI tools • Use specific criteria to assess their strengths and limitations EDCI 59100-016 Week 3 Lecture Notes • Decide how (or whether) these tools should support your search, screening, and evaluation work This activity will help you use AI more critically and support stronger decision-making in your literature review plan and assignments. To do this effectively, you first need to understand how AI systems generate and prioritize information. AI can shape what you see — so you need verification + triangulation.

AI and Information Challenges

• Bias • Hallucinations & Misinformation • The "Black Box" problem • Copyright and legal issues • Privacy and data protection • Research quality concerns • Environmental considerations • Social, cultural, and ethical impacts These challenges are not abstract. They directly affect what you see when you use AI tools. In the next sections, you will learn how these issues emerge during AI search and content generation. For example, an AI tool may confidently provide incomplete or misleading summaries, or prioritize certain perspectives without making that process visible to you

Why This Matters for Your Literature Review Process

As you begin searching, screening, and evaluating literature, you may also use AI tools (e.g., ChatGPT, Copilot, or AI-enabled search tools) to: • generate keywords • summarize articles • suggest sources • explain concepts However, AI does not search or "think" like a researcher. Understanding how AI works will help you: • use it more effectively • recognize its limitations • avoid relying on inaccurate or incomplete information How AI "Searches" and Retrieves Information AI search process: AI tools often follow a process like this: • Input → You provide a prompt, question, or keywords • Interpretation → AI interprets what you are asking • Matching / Retrieval → It identifies related information (from training data or connected sources) EDCI 59100-016 Week 3 Lecture Notes • Ranking → It prioritizes what seems most relevant • Output → It generates a response (text, summary, suggestions)

Important takeaway

What AI tools do NOT do: AI does NOT: • systematically search databases like ERIC or Scopus • apply inclusion/exclusion criteria • verify the credibility of sources Instead, it generates outputs based on patterns and probabilities

How AI Generates Content

When AI produces text, it is: predicting likely sequences of words based on patterns What this means for your work as a researcher: As a result, AI can: • summarize and explain information • generate readable text quickly • suggest connections between ideas Caution: But it can also: • oversimplify complex research • omit important details • combine ideas incorrectly • generate sources that sound real but are not • present uncertain information confidently

What This Means for Your Literature Review

AI can help you:

• brainstorm keywords and synonyms • explore related concepts • clarify unfamiliar terminology • generate initial summaries (to verify later) In your own work, this means using AI to support early-stage tasks (e.g., generating keywords), while continuing to rely on structured processes such as database searching, screening criteria, and evaluation frameworks. Caution: AI should NOT replace: • database searching • screening sources • applying inclusion/exclusion criteria • evaluating source quality • identifying valid patterns across studies These are your responsibilities as the researcher Example: An AI tool might generate a convincing summary of "research on student motivation" and include citations that look real—but when you search for those articles, they do not exist or do not match the claims. This is why verification is essential.

Key Idea

AI can support your thinking—but it does not replace your judgment.

Evaluating AI Tools (Key Criteria)

The most important criteria will vary depending on how you are using the tool. For this activity, focus on a few key areas: • Transparency: Where does the information come from... • Accuracy / Verifiability: Can you confirm the claims or sources... • Bias / Framing: Does the response privilege certain perspectives... • Search Strategy Support: Does it help generate useful keywords... • Evaluation Support: Does it help determine relevance of sources... • Limitations / Risks: What could mislead a researcher... A more detailed framework is available here: Evaluating AI Tools (Purdue)

Expanded Criteria for Evaluating AI Tools for Literature Review Work

The criteria above provide a quick starting point. The expanded framework below helps you evaluate tools more deeply if needed.

1. Purpose and Scope • What is the tool designed to do... • Is it designed for general use, academic research, or literature searching...

1. Transparency • Does the tool explain where its information comes from... • Does it indicate whether it retrieves actual sources, generates text, or both...

1. Accuracy / Verifiability • Can you verify the information it gives... • Does it provide citations, links, or identifiable sources...

1. Bias / Framing • Does the output appear balanced... • Does it overemphasize certain ideas,

outcomes, or perspectives... • Does it flatten differences across studies...

1. Usefulness for Search Strategy • Does the tool generate useful keywords, synonyms, or related concepts... • Does it help refine a search in meaningful ways...

1. Usefulness for Screening / Evaluation • Can the output help you judge relevance... • Or does it oversimplify what sources are actually about...

1. Limitations / Risks • What could mislead a researcher if they relied on the tool too heavily... • What still requires human judgment...

Now that you understand how AI systems generate and prioritize information, you will apply these ideas by evaluating AI tools more systematically. In the activity below, you will evaluate AI tools using specific criteria and examine how they influence literature review searching, screening, and evaluation.

Activity: Evaluate AI Tools for Literature Review Searching

Purpose: Evaluate how AI tools retrieve, generate, and present information, and what that means for your literature review work.

What You Will Do

• Select three AI tools • Choose 1–2 evaluation criteria • Evaluate each tool • Document and reflect on your findings

Step 1: Choose AI Tools

• ChatGPT • Microsoft Copilot • Perplexity • Elicit • Semantic Scholar • Scite • Consensus • Research Rabbit You may also evaluate another tool you use.

Step 2: Select Criteria

Choose 1–2 criteria from the list above that you will use to evaluate each tool.

Step 3: Evaluate Each Tool

Use one or both approaches:

Review Tool Information

• About page • Documentation • Help / FAQ

Test the Tool

Example prompts: • Keyword generation: “Suggest search terms for my topic: [your topic]” • Source explanation: “Explain the research on [topic] and provide sources” • Pattern identification: “What patterns appear in the literature on [topic]...” Focus on what the tool produces, not just what it claims to do. Step 4: Document Your Findings Record: • What the tool claims • What it actually does • Strengths and limitations • How you would use (or not use) the tool Use the worksheet to document your evaluation as you examine each tool: Download AI Tool Evaluation Matrix (Excel) Step 5: Reflection Based on your evaluation, how might this tool’s strengths, limitations, or potential bias influence your decision to use it in your research... Strong responses: • connect your evaluation findings to specific tool behaviors • explain how these influence your research decisions • describe how you will use (or limit use of) AI tools moving forward The goal is not just to use AI tools, but to use them critically and intentionally as part of your research process.